

Ex-32 - Eight circuit

1) As for ex 31, we will use some transformation matrix to move ~~a point p~~ ^{the vector field} to field.
For each point p of the field:

We use $g(p) = \begin{pmatrix} \rho & 0 \\ 0 & \rho \cdot \varepsilon \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \varepsilon \end{pmatrix} p + c$ — new center of the attractor
 $\swarrow$ multiply by $\rho \cdot \text{Id}$ $\downarrow$ ε control the clockwise
 to change the amplitude,
 so the radius

$$g(p) = Dp + c \text{ with } D = \begin{pmatrix} \rho & 0 \\ 0 & \rho \cdot \varepsilon \end{pmatrix}$$

To use it on our vector field ϕ_0 , we have to do $\phi_{\varepsilon, \rho, c} = g \circ \phi_0$
 but from ex 31, we have $(g \circ f)(x) = A \cdot f(A^{-1}x)$

Then: $\phi_{\varepsilon, \rho, c}(p) = g \circ \phi_0(p) = D \cdot \phi_0 \cdot D^{-1}(p - c)$

2) We want to follow the vector field describe by $\phi = \phi_{\varepsilon, \rho, c}$

We can only control the orientation speed of the robot. So we want it follows $\Psi = \text{angle}(\phi)$

Let's called the error of angle $e_\Psi = x_3 - \Psi$

So $\dot{e}_\Psi = \dot{x}_3 - \dot{\Psi} = u - \dot{\Psi}$

$\Psi = \text{Arctan}\left(\frac{\phi_2}{\phi_1}\right)$ where $\phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$

$$\dot{\Psi} = \frac{1}{1 + \left(\frac{\phi_2}{\phi_1}\right)^2} \times \frac{d}{dt} \left(\frac{\phi_2}{\phi_1} \right) = \frac{1}{1 + \left(\frac{\phi_2}{\phi_1}\right)^2} \times \frac{\phi_2' \phi_1 - \phi_2 \phi_1'}{\phi_1^2} = \frac{\phi_2' \phi_1 - \phi_2 \phi_1'}{\phi_1^2 + \phi_2^2}$$

Knowing $\phi(p) = D \phi_0 D^{-1}(p - c)$, we deduce:

$$\begin{aligned} \dot{\phi}(p) = \begin{pmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{pmatrix} &= \frac{d}{dt} \left(D \phi_0 D^{-1}(p - c) \right) = D \cdot \frac{d\phi_0}{dp} (D^{-1}(p - c)) \times \frac{d}{dt} (D^{-1}(p - c)) \\ &= D \cdot \frac{d\phi_0}{dp} [D^{-1}(p - c)] \cdot D^{-1} \cdot \dot{p} \end{aligned}$$

We know $\dot{p} = \begin{pmatrix} \cos x_3 \\ \sin x_3 \end{pmatrix}$

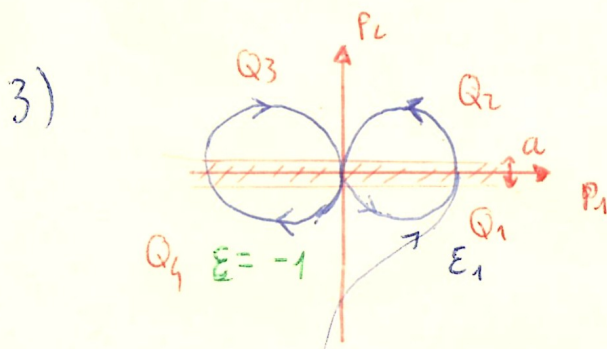
$$\frac{d\Phi_0}{dP} = \begin{pmatrix} \frac{\partial \Phi_0}{\partial P_1} & \frac{\partial \Phi_0}{\partial P_2} \end{pmatrix} = \begin{pmatrix} -3p_1^2 - p_2^2 + 1 & -2p_1p_2 - 1 \\ -2p_1p_2 + 1 & -3p_2^2 - p_1^2 + 1 \end{pmatrix}$$

We can now express Ψ and $\dot{\Psi}$ as

We choose $u = \dot{\Psi} - v$ to get $\dot{e}_\Psi = v$

Then $v = -e_\Psi$ in order to have a control feedback with one pole to -1 .

We could also surround e_Ψ and u by a sawtooth function to have a 2π period when the robot is turning.



We will use an automation with 4 states to change the behavior of the robot with the change of vector fields based on p_1, p_2 values.

Like if the robot is in a specific state Q_i and an event like $p_2 > a$ then the robot will change q_i and we will be in the Q_{i+1} space.